

SINGULAR OPTIMAL STOCHASTIC CONTROLS II: DYNAMIC PROGRAMMING*

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Abstract. The dynamic programming principle for a multidimensional singular stochastic control problem is established in this paper. When assuming Lipschitz continuity on the data, it is shown that the value function is continuous and is the unique viscosity solution of the corresponding Hamilton–Jacobi–Bellman equation.

Key words. singular controls, control rules, value function, dynamic programming principle, Hamilton–Jacobi–Bellman equation, viscosity solution

AMS subject classifications. 49J30, 49A55, 60G44, 93E20

1. Introduction. In [8] we applied a direct method to study the existence of optimal controls for the stochastic control problem in which the state is governed by the stochastic differential equation

$$x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + \int_s^t g(\theta) dv_\theta$$

on some filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, where $b(\cdot, \cdot, \cdot)$, $\sigma(\cdot, \cdot, \cdot)$, $g(\cdot)$ are given deterministic functions, $(B_t, t \geq 0)$ is a d -dimensional Brownian motion (in fact, B need not be d -dimensional), x is the initial state at time s , and $u : [0, T] \mapsto U$, $v : [0, T] \mapsto \mathbb{R}^k$, with v nondecreasing componentwise, stand for the controls.

The expected cost has the form

$$J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, u_t) dt + \int_{[s, T)} c(t) \cdot dv_t \right\},$$

where $f(\cdot, \cdot, \cdot) : [0, T] \times \mathbb{R}^d \times U \mapsto \mathbb{R}$, $c(\cdot) : [0, T] \mapsto \mathbb{R}_+^k$ are given. We assume that the cost of applying the singular control is positive, i.e., $c^i(\cdot) > 0$, $i = 1, \dots, k$. For this type of problem, the reader may consult the paper by Haussmann and Suo [8] and the list of references therein.

This paper is a continuation of Haussmann and Suo [8]. As is well known for the classical stochastic control problem, the dynamic programming principle is satisfied and, if the value function has appropriate regularity, it satisfies a second-order nonlinear partial differential equation (the Hamilton–Jacobi–Bellman equation) (cf. Fleming and Rishel [4] and Lions [10], among others). This is still the case for singular stochastic control where the Hamilton–Jacobi–Bellman equation is a second-order variational inequality (see Fleming and Soner [5] and the list of references in Haussmann and Suo [8]). In this paper, in §3 we adopt a probabilistic approach used in Haussmann [6], Haussmann and Lepeltier [7], and El Karoui, Nguyen, and Jeanblanc-Piqué [3] to establish the dynamic programming principle under very mild conditions of the data. Then, in §4, assuming Lipschitz continuity of the coefficients, we prove

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that the value function is continuous. In §5 the Hamilton–Jacobi–Bellman equation is derived heuristically, and the value function is shown to be the unique viscosity solution of this equation. For the reader’s convenience, the main results of Haussmann and Suo [8] are recalled in §2, along with the formulation of the problem.

We list some notation that will be used throughout this paper:

- $\mathbb{R}^d, \mathbb{R}$ denote the d -dimensional Euclidean space and the real line, respectively. $\mathbb{R}_+ = \{x \in \mathbb{R}, x \geq 0\}$ and $\mathbb{R}_+^d$ is defined similarly. For $x = (x^i), y = (y^i) \in \mathbb{R}^d$, $x \cdot y = \sum_{i=1}^d x^i y^i$.
- $T > 0$ is the fixed horizon, and $\Sigma = [0, T] \times \mathbb{R}^d$.
- $\mathcal{D}^d[0, T]$ denotes the collection of $\mathbb{R}^d$ -valued functions defined on $[0, T]$ which are left continuous and have right limits.
- $\mathcal{A}^k[0, T]$ denotes the collection of functions $a : [0, T] \mapsto \mathbb{R}_+^k$ such that $a = (a^i) \in \mathcal{D}^k[0, T]$ and a^i is nondecreasing with $a^i(0) = 0$, $i = 1, \dots, k$.
- $\mathcal{S}^{l \times k}$ is the space of $l \times k$ matrices with the $l \times k$ -dimensional Euclidean norm.
- If Y is a metric space, $\mathcal{B}(Y)$ denotes the corresponding Borel σ -field, and $f \in \mathcal{B}(Y)$ means that f is a $\mathcal{B}(Y)$ -measurable real-valued function. We denote by $\mathcal{M}_1(Y)$ ($\mathcal{M}_+(Y)$, respectively) the space of probabilities (nonnegative Radon measures, respectively) on Y with the weak convergence topology.
- U , called the control set, is a compact metric space. It is well known that $\mathcal{M}_1(U)$ is also a compact metrizable space. If $\phi : U \mapsto \mathbb{R}$ is a bounded measurable function, we can extend ϕ to $\mathcal{M}_1(U)$ by letting

$$\phi(\mu) \equiv \int_U \phi(u) \mu(du).$$

Define

$$\mathcal{U} \equiv \{\mu : [0, T] \mapsto \mathcal{M}_1(U) \text{ is Borel measurable}\}.$$

- If X is a random variable on a probability space $(\Omega, \mathcal{F}, P)$, the expectations of X will be denoted by $E^P(X)$. $\mathcal{M}_2^c$ is the family of continuous square integrable martingales on some given probability space $(\Omega, \mathcal{F}, P)$ with a filtration $\{\mathcal{F}_t\}$.
- C stands for a constant, but not necessarily the same one from line to line.

2. Formulation of the problem. We consider the following optimal control problem in which we allow both classical control and singular control to act at the same time, i.e., the dynamics are in the form,

$$(2.1) \quad x_t = x + \int_s^t b(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, \mu_\theta) dB_\theta + \int_s^t g(\theta) dv_\theta \quad \text{a.s.}$$

for $(t, x) \in \Sigma$, $s \leq t \leq T$, where

- $(\sigma, b) : \Sigma \times U \mapsto \mathcal{S}^{d \times d} \times \mathbb{R}^d$, $g : [0, T] \mapsto \mathcal{S}^{d \times k}$,
- $(B_t, 0 \leq t \leq T)$ is a d -dimensional Brownian motion on some probability space, and
- $v \in \mathcal{A}^k[0, T]$.

We assume that σ, b are bounded measurable functions, continuous with respect to (x, u) , and g is continuous on $[0, T]$.

DEFINITION 2.1. A relaxed control is a term $\alpha = (\Omega, \mathcal{F}, P, \mathcal{F}_t, B_t, x_t, \mu_t, v_t, s, x)$ such that

- (1) $(s, x) \in \Sigma$;
- (2) $(\Omega, \mathcal{F}, P)$ is a probability space with the filtration $\{\mathcal{F}_t\}_{t \geq 0}$;
- (3) $\mu \in \mathcal{U}$ is progressively measurable with respect to $\mathcal{F}_t$;
- (4) v is an $\mathbb{R}_+^k$ -valued process progressively measurable with respect to $\mathcal{F}_t$; the sample paths of v are in $\mathcal{A}^k[0, T]$, i.e., for each $\omega \in \Omega$, $v(\omega) \in \mathcal{A}^k[0, T]$;
- (5) B_t is a standard d -dimensional Brownian motion on $(\Omega, \mathcal{F}, P, \mathcal{F}_t)$ and x_t , the state process, is $\mathcal{F}_t$ -adapted with sample paths in $\mathcal{D}^d[0, T]$ such that (2.1) is satisfied. We assume that $x_r = x$ for $0 \leq r \leq s$.

We call (s, x) the initial condition of the relaxed control α .

The collection of relaxed controls with initial condition (s, x) is denoted by $\tilde{\Lambda}_{s,x}$. For $\alpha \in \tilde{\Lambda}_{s,x}$, the associated cost is defined as follows:

$$(2.2) \quad J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, \mu_t) dt + \int_{[s,T)} c(t) \cdot dv_t \right\},$$

where

- $f : \Sigma \times U \mapsto \mathbb{R}$ is a measurable function and is lower semicontinuous in (x, u) , satisfying

$$-K \leq f(t, x, u) \leq C(1 + \|x\|^m), \quad (t, x, u) \in \Sigma \times U$$

for some constants $m \geq 0$, $K \geq 0$ and $C \geq 0$;

- $c = (c^i) : [0, T] \mapsto \mathbb{R}^k$ is lower semicontinuous and $c^i > 0$, $1 \leq i \leq k$.

The value function is defined by

$$(2.3) \quad W(s, x) = \inf_{\alpha \in \tilde{\Lambda}_{s,x}} J(\alpha).$$

A relaxed control $\alpha^* \in \tilde{\Lambda}_{s,x}$ is called an optimal relaxed control if $W(s, x) = J(\alpha^*)$.

Throughout this paper we write

$$\int_s^t k(\theta) \cdot da(\theta) = \sum_{i=1}^d \int_{[s,t)} k_i(\theta) da_i(\theta)$$

for any $\mathbb{R}^k$ -valued Borel-measurable function $k = (k_i)$ and $a = (a_i) \in \mathcal{A}^k[0, T]$.

Now we introduce the canonical space for our problem. Define

$$\Omega \equiv \mathcal{D}^d[0, T] \times \mathcal{U} \times \mathcal{A}^k[0, T],$$

where $\mathcal{D}^d[0, T]$, $\mathcal{U}$ and $\mathcal{A}^k[0, T]$ are endowed with the pseudopath topology, stable topology, and weak convergence topology, respectively. Ω is metrizable and separable under the product topology; see Haussmann and Suo [8] for details. Let $\tilde{\mathcal{D}}, \tilde{\mathcal{U}}, \tilde{\mathcal{A}}$ denote their Borel σ -fields, $\tilde{\mathcal{D}}_t, \tilde{\mathcal{U}}_t, \tilde{\mathcal{A}}_t$, the σ -fields up to time t , and define

$$\begin{aligned} \mathcal{F} &\equiv \tilde{\mathcal{D}} \times \tilde{\mathcal{U}} \times \tilde{\mathcal{A}}, \\ \mathcal{F}_t &\equiv \tilde{\mathcal{D}}_t \times \tilde{\mathcal{U}}_t \times \tilde{\mathcal{A}}_t. \end{aligned}$$

Let $a = \sigma\sigma^*$, where $*$ denotes the transpose of a matrix, and define

$$\mathcal{L} \equiv \frac{1}{2} \sum_{i,j} a_{ij} \frac{\partial^2}{\partial x_i \partial x_j} + \sum_i b_i \frac{\partial}{\partial x_i}.$$

We recall the following definition from Haussmann and Suo [8]

DEFINITION 2.2. A control rule is a probability R on the measurable space $(\Omega, \mathcal{F})$ such that

$$\tilde{\alpha} = (\Omega, \mathcal{F}, \mathcal{F}_t, R, x_t, \mu_t, v_t, s, x) \in \tilde{\Lambda}_{s,x},$$

where

$$x_t(\omega) = x_t, \mu_t(\omega) = \mu_t, v_t = v_t(\omega)$$

for $\omega = (x, \mu, v) \in \Omega$, i.e.,

(1) $(\Omega, \mathcal{F}, R)$ is a probability space with the filtration $\mathcal{F}_t$, $(s, x) \in \Sigma$, and

$$R(x_r = x, \mu_r = \delta^0, v_r = 0, 0 \leq r \leq s) = 1,$$

where δ^0 denotes the Dirac measure at an arbitrary but fixed point $u^0 \in U$;

(2) $\forall \phi \in C_b^2(\mathbb{R}^d)$, $M_t \phi$ ($s \leq t \leq T$) is in $\mathcal{M}_2^c$ on the filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t, R)$ ($s \leq t \leq T$), where

$$(2.4) \quad \begin{aligned} M_t \phi(\omega) \equiv & \phi(x_t) - \int_s^t \mathcal{L} \phi(\theta, x_\theta, \mu_\theta) d\theta - \int_s^t \nabla_x \phi(x_\theta) \cdot g(\theta) dv_\theta \\ & - \sum_{s \leq \theta < t} [\phi(x_{\theta+}) - \phi(x_\theta) - \nabla_x \phi(x_\theta) \cdot \Delta x_\theta]. \end{aligned}$$

We write $J(\alpha) = J(s, R)$.

Define

$$(2.5) \quad \Gamma_s(\omega) \equiv \int_s^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta$$

for $\omega = (x, \mu, v)$. Then, by definition, $J(\alpha) = E^R \Gamma_s$. We denote by $\mathcal{R}_{s,x}$ the space of control rules with initial condition (s, x) . We will suppress s in $J(s, R)$ when it is clear that $R \in \mathcal{R}_{s,x}$. It is shown in Haussmann and Suo [8] that for any relaxed control $\alpha \in \tilde{\Lambda}_{s,x}$, there exists a control rule $R \in \mathcal{R}_{s,x}$ such that $J(R) = J(\alpha)$. Thus the control problem can be described in terms of control rules.

We recall the main results from Haussmann and Suo [8].

THEOREM 2.3 (Haussmann and Suo [8]). (a) For each $(s, x) \in \Sigma$ there exists an optimal control rule $P^* \in \mathcal{R}_{s,x}$, i.e., $W(s, x) = J(s, P^*)$.

(b) The value function $W(\cdot, \cdot)$ is Borel measurable on Σ .

(c) There exists a Borel-measurable map $H(\cdot, \cdot) : \Sigma \mapsto \mathcal{M}_1(\Omega)$ such that for each $(s, x) \in \Sigma$, $H(s, x) \in \mathcal{R}_{s,x}^o$, where

$$\mathcal{R}_{s,x}^o = \{P \in \mathcal{R}_{s,x}, J(s, P) = W(s, x)\}.$$

We call H a measurable selector of $\mathcal{R}^o$.

3. The dynamic programming principle. In this section we will apply the method used in Haussmann [6], Haussmann and Lepeltier [7], and El Karoui, Nguyen, and Jeanblanc-Picqué [3] to establish the dynamic programming principle. Note that this method does not require any regularity of the value function W .

3.1. Some preparations. We denote by $\mathcal{F}^s$ the σ -algebra of events that occur after time s , i.e., $\mathcal{F}^s = \sigma\{\omega_t, s \leq t \leq T\}$.

Define $\Theta_{s,\bar{\omega}} : \Omega \mapsto \Omega$ by

$$(3.1) \quad \Theta_{s,\bar{\omega}}(\omega)_t = \begin{cases} \bar{\omega}_t, & 0 \leq t \leq s, \\ (x_t, \mu_t, \bar{v}_s + v_t - v_s), & s < t \leq T. \end{cases}$$

Note that if $\bar{\omega} = \Theta_{s,\bar{\omega}}(\omega)$, then $\bar{\omega} = \bar{\omega}$ on $[0, s]$, $(x.(\bar{\omega}), \mu.(\bar{\omega})) = (x.(\omega), \mu.(\omega))$ on $(s, T]$, and $v.(\bar{\omega}) - v_s(\bar{\omega}) = v.(\omega) - v_s(\omega)$ on $[s, T]$.

LEMMA 3.1. *If P is a probability on $(\Omega, \mathcal{F}^s)$ ($0 \leq s \leq T$) and $\bar{\omega} \in \Omega$, then there exists a unique probability measure, denoted by $\delta_{\bar{\omega}} \otimes_s P$ on $(\Omega, \mathcal{F})$ such that*

$$(3.2) \quad \delta_{\bar{\omega}} \otimes_s P(\omega : \omega_t = \bar{\omega}_t, 0 \leq t \leq s) = 1,$$

$$(3.3) \quad \delta_{\bar{\omega}} \otimes_s P(A) = P(\Theta_{s,\bar{\omega}}^{-1}(A)) \quad \forall A \in \mathcal{F}^s,$$

where by $\omega_t = \bar{\omega}_t$, $0 \leq t \leq s$, we mean $x_t = \bar{x}_t$, $v_t = \bar{v}_t$, $0 \leq t \leq s$, and $\mu_t = \bar{\mu}_t$ a.e. on $[0, s]$.

Proof. The uniqueness of such a probability measure is obvious, so we only need to show its existence.

If I is a subinterval of $[0, T]$ we write $V(I)$ for the set of measurable functions

$$I \mapsto \mathbb{M}_1(U).$$

Let us recall an equivalent definition of the stable topology on $\mathcal{U} = V([0, T])$. For $\mu \in \mathcal{U}$ define a mapping $i : \mathcal{U} \mapsto C([0, T], \mathbb{M}_+(U))$ by

$$i(\mu)(\cdot) = \int_0^\cdot \mu_\theta d\theta.$$

The topology on $\mathcal{U}$ induced by i is exactly the stable topology we introduced in Haussmann and Suo [8]. For a discussion of this see Haussmann and Lepeltier [7, §3.10]. Similarly, we can consider the topology on $V(I)$ induced by the mapping

$$i_I(\mu)(\cdot) = \int_0^\cdot \mathbf{1}_I(\theta) \mu_\theta d\theta \in C([0, T], \mathbb{M}_+(U)), \quad \mu \in V(I).$$

Write $\mathcal{A}_I^k$ for the $\mathbb{R}_+^k$ -valued nondecreasing functions on I with the inherited topology from $\mathcal{A}^k[0, T]$. Let

$$\begin{aligned} X_0 &\equiv \mathcal{D}^d[0, s] \times i_{[0,s]} V([0, s]) \times \mathcal{A}_{[0,s]}^k, \\ X &\equiv \mathcal{D}^d[s, T] \times i_{[s,T]} V([s, T]) \times \mathcal{A}_{[s,T]}^k, \\ \tilde{X} &\equiv X_0 \times X, \end{aligned}$$

and define $\Phi_0 : \Omega \mapsto X_0$, $\Phi : \Omega \mapsto X$ by

$$\begin{aligned} \Phi_0(\omega)_t &= \left(x_t, \int_0^t \mu_\theta d\theta, v_t \right), \quad 0 \leq t \leq s, \\ \Phi(\omega)_{t'} &= \left(x_{t'}, \int_s^{t'} \mu_\theta d\theta, v_{t'} \right), \quad s \leq t' \leq T, \end{aligned}$$

where $\omega = (x, \mu, v) \in \Omega$. Define $\Psi : \tilde{X} \mapsto \Omega$ by

$$\Psi(\omega_1, \omega_2) = \Theta_{s, \tau_0(\omega_1)}(\tau(\omega_2)),$$

where $\tau_0 : X_0 \mapsto \Omega$, $\tau : X \mapsto \Omega$:

$$\begin{aligned}\tau_0(\tilde{\omega})_t &= (\tilde{x}_{t \wedge s}, i_{[0, s]}^{-1}(\tilde{\mu})(t \wedge s), \tilde{v}_{t \wedge s}), \\ \tau(\omega)_t &= (x_{t \vee s}, i_{[s, t]}^{-1}(\mu)(t \vee s), v_{t \vee s})\end{aligned}$$

with $\tilde{\omega} = (\tilde{x}, \tilde{\mu}, \tilde{v}) \in X_0$, $\omega = (x, \mu, v) \in X$.

We define a probability $\tilde{P}$ on $\tilde{X}$ by $\tilde{P} = \delta_{\tilde{\omega}} \circ \Phi_0^{-1} \times P \circ \Phi^{-1}$, and let

$$\bar{P} = \tilde{P} \circ \Psi^{-1}.$$

Now we verify that $\bar{P}$ satisfies conditions (3.2) and (3.3). Note that $\tau_0 \circ \Phi_0(\bar{\omega})_t = \bar{\omega}_t$, $0 \leq t \leq s$, so we have

$$\begin{aligned}\bar{P}(\omega : \omega_t = \bar{\omega}_t, 0 \leq t \leq s) &= \tilde{P}((\omega_1, \omega_2) : \Theta_{s, \tau_0(\omega_1)}(\tau(\omega_2))_t = \bar{\omega}_t, 0 \leq t \leq s) \\ &= \tilde{P}((\omega_1, \omega_2) : \tau_0(\omega_1)_t = \bar{\omega}_t, 0 \leq t \leq s) \\ &= \delta_{\tilde{\omega}}(\omega_1 : \tau_0 \circ \Phi_0(\omega_1)_t = \bar{\omega}_t, 0 \leq t \leq s) \\ &= 1.\end{aligned}$$

For $A \in \mathcal{F}^s$,

$$\begin{aligned}\bar{P}(A) &= \tilde{P}((\omega_1, \omega_2) : \Psi(\omega_1, \omega_2) \in A) \\ &= \tilde{P}((\omega_1, \omega_2) : \Theta_{s, \tau_0(\omega_1)}(\tau(\omega_2)) \in A) \\ &= \tilde{P}((\omega_1, \omega_2) : \tau(\omega_2) \in \Theta_{s, \tau_0(\omega_1)}^{-1}(A)) \\ &= \int_{X_0} P \circ \Phi^{-1}(\omega_2 : \tau(\omega_2) \in \Theta_{s, \tau_0(\omega_1)}^{-1}(A)) \delta_{\tilde{\omega}} \circ \Phi_0^{-1}(d\omega_1) \\ &= \int_{X_0} P(\omega : \tau \circ \Phi(\omega) \in \Theta_{s, \tau_0(\omega_1)}^{-1}(A)) \delta_{\Phi_0(\tilde{\omega})}(d\omega_1) \\ &= P(\omega : \tau \circ \Phi(\omega) \in \Theta_{s, \tau_0(\Phi_0(\tilde{\omega}))}^{-1}(A)) \\ &= P(\Theta_{s, \tilde{\omega}}^{-1}(A)),\end{aligned}$$

since $\tau \circ \Phi(\omega)_t = \omega_t$ holds on $[s, T]$. The lemma is proved by letting $\delta_{\tilde{\omega}} \otimes_s P = \bar{P}$. $\square$

Remark 3.2. Note that for $P \in \mathcal{R}_{s, x}$, there exists a P -null set N_0 such that if $\bar{\omega} \notin N_0$, $A \in \mathcal{F}^s$, then $P(A) = P \circ \Theta_{s, \bar{\omega}}^{-1}(A)$ and

$$E^P \left\{ \int_s^t h(\theta) \cdot dv_\theta \right\} = E^{P \circ \Theta_{s, \bar{\omega}}^{-1}} \left\{ \int_s^t h(\theta) \cdot dv_\theta \right\}, \quad t \geq s$$

for any bounded $\mathbb{R}^k$ -valued Borel-measurable function $h(\cdot)$. These properties will be used repeatedly in the rest of this section.

Assume that τ is an $\mathcal{F}_t$ -stopping time, $0 \leq \tau \leq T$. A τ -transition probability (τ -t.p.) is a family $\{Q_\omega : \omega \in \Omega\}$ of probability measures on $(\Omega, \mathcal{F})$ such that

$$\omega \mapsto Q_\omega(A) \text{ is } \mathcal{F}_\tau \text{ measurable } \forall A \in \mathcal{F}.$$

Note that $\mathcal{F}_\tau$ is the collection of sets A such that $A \cap \{\tau \leq t\} \in \mathcal{F}_t \forall t \leq T$. For a fixed τ and ω such that $\tau(\omega) \leq T$, we denote by $\mathcal{F}^{\tau(\omega)}$ the σ -field generated by the sets of the form

$$(3.4) \quad \left\{ \tilde{\omega} \in \Omega : \tilde{x}_t \in A, \int_{\tau(\omega)}^t \tilde{\mu}_\theta d\theta \in B, \tilde{v}_t \in C \right\},$$

where $A \in \mathcal{B}(\mathbb{R}^d)$, $B \in \mathcal{B}(M_+(U))$, $C \in \mathcal{B}(\mathbb{R}^k)$, $\tau(\omega) \leq t \leq T$, so $\mathcal{F}^{\tau(\omega)}$ is the collection of events that occur after the time $\tau(\omega)$. Note that the topology on Ω is separable, so $\mathcal{F}_t$ is countably generated, and for a given probability P on $(\Omega, \mathcal{F})$, the regular conditional probability distribution (r.c.p.d.) of P for given $\mathcal{F}_\tau$ exists and will be denoted by $P_{\tau, \omega}$.

Given a stopping time τ ($0 \leq \tau \leq T$) and a τ -t.p. Q_ω , if $\bar{\omega} \in \Omega$, then from Lemma 3.1 we know that there exists a unique $\delta_{\bar{\omega}} \otimes_\tau Q_\omega \in M_1(\Omega)$ such that

$$\begin{aligned} \delta_{\bar{\omega}} \otimes_\tau Q_\omega \{ \tilde{\omega} : \tilde{\omega}_t = \bar{\omega}_t, 0 \leq t \leq \tau(\omega) \} &= 1, \\ \delta_{\bar{\omega}} \otimes_\tau Q_\omega(A) &= Q_\omega(\Theta_{\tau, \bar{\omega}}^{-1}(A)) \quad \forall A \in \mathcal{F}^{\tau(\omega)}. \end{aligned}$$

We write $\delta_{\bar{\omega}} \otimes_\tau Q_\omega = Q_{\bar{\omega}}^\tau$. When $\bar{\omega} = (x(\tau(\omega)), \delta^0, 0)$, we write

$$\delta_{\bar{\omega}} \otimes_\tau Q_\omega = \overset{\circ}{Q}_{\omega}^\tau.$$

It can be seen easily that for $s \leq T$ and a stopping time $\tau : s \leq \tau \leq T$,

$$(3.5) \quad Q_\omega(\Gamma_{\tau(\omega)}) = Q_\omega^\tau(\Gamma_{\tau(\omega)}) = \overset{\circ}{Q}_{\omega}^\tau(\Gamma_{\tau(\omega)}),$$

where Γ is defined by (2.5).

If $P \in M_1(\Omega)$, τ is a stopping time, and $\{Q_\omega\}$ is a τ -t.p., then we have the following result, which is analogous to Theorem 6.1.2 in Stroock and Varadhan [11].

LEMMA 3.3. *There exists a unique probability, denoted by $P \otimes_\tau Q$, such that*

- (1) $P \otimes_\tau Q(A) = P(A)$ if $A \in \mathcal{F}_\tau$,
- (2) *the r.c.p.d. of $P \otimes_\tau Q$ with respect to $\mathcal{F}_\tau$ is Q_ω^τ .*

Proof. The proof of [11, Thm. 6.1.2] can be adopted here with minor modifications (cf. Suo [12]). $\square$

LEMMA 3.4. *Assume $P \in \mathcal{R}_{s,x}$, τ is an $\mathcal{F}_t$ -stopping time: $s \leq \tau \leq T$, and $\bar{\omega} \in \Omega$. Then $(M_t\phi, \mathcal{F}_t, P \circ \Theta_{\tau, \bar{\omega}}^{-1})$ is a martingale after $\tau_{\bar{\omega}}$ for $\phi \in C_b^2(\mathbb{R}^d)$.*

Proof. Since $P \in \mathcal{R}_{s,x}$, we know that $(M_t\phi, \mathcal{F}_t, P)$ is a martingale, i.e.,

$$(3.6) \quad \int_A M_t\phi dP = \int_A M_u\phi dP \quad \forall A \in \mathcal{F}_u, \quad s \leq u < t \leq T,$$

or

$$(3.7) \quad E^P \{ 1_A(\cdot) [M_t\phi(\cdot) - M_u\phi(\cdot)] \} = 0.$$

When $t \geq \tau_{\bar{\omega}}$, we have

$$\begin{aligned} M_t\phi(\Theta_{\tau, \bar{\omega}}(\omega)) &= \phi(x_t) - \int_{\tau_{\bar{\omega}}}^t \mathcal{L}\phi(\theta, x_\theta, \mu_\theta) d\theta - \int_{\tau_{\bar{\omega}}}^t \nabla_x \phi(x_\theta) \cdot g(\theta) dv_\theta^c \\ &\quad - \sum_{\tau_{\bar{\omega}} < \theta < t} [\phi(x_{\theta+}) - \phi(x_\theta)] - [\phi(x_{\tau+}) - \phi(\bar{x}_\tau)] \\ &\quad - \int_s^{\tau_{\bar{\omega}}} \mathcal{L}\phi(\theta, \bar{x}_\theta, \bar{\mu}_\theta) d\theta - \int_s^{\tau_{\bar{\omega}}} \nabla_x \phi(\bar{x}_\theta) \cdot g(\theta) d\bar{v}_\theta^c \\ &\quad - \sum_{s \leq \theta < \tau_{\bar{\omega}}} [\phi(\bar{x}_{\theta+}) - \phi(\bar{x}_\theta)]. \end{aligned}$$

Therefore we can get for $\tau_{\bar{\omega}} \leq u \leq t \leq T$, $A \in \mathcal{F}_u$,

$$\begin{aligned} & E^{P \circ \Theta_{\tau_{\bar{\omega}}, \bar{\omega}}^{-1}} \{1_A(\cdot) [M_t \phi(\cdot) - M_u \phi(\cdot)]\} \\ &= E^P \{1_A(\Theta_{\tau_{\bar{\omega}}, \bar{\omega}}(\cdot)) [M_t \phi(\Theta_{\tau_{\bar{\omega}}, \bar{\omega}}(\cdot)) - M_u \phi(\Theta_{\tau_{\bar{\omega}}, \bar{\omega}}(\cdot))]\} \\ &= E^P \{1_A(\Theta_{\tau_{\bar{\omega}}, \bar{\omega}}(\cdot)) [M_t \phi(\cdot) - M_u \phi(\cdot)]\} \\ &= 0, \end{aligned}$$

because it is obvious that $\Theta_{\tau_{\bar{\omega}}, \bar{\omega}}^{-1}(A) \in \mathcal{F}_u$. The lemma is thus proved. $\square$

Recall that for a stopping time τ , $P_{\tau, \omega}$ denotes the regular conditional probability distribution of P given $\mathcal{F}_\tau$. $P_{\tau, \omega}$ is obviously a τ -t.p., so $\overset{\circ}{P}_\omega^\tau \equiv \delta_{\bar{\omega}} \otimes_\tau P_{\tau, \omega}$ is defined, where $\bar{\omega} = (x(\tau(\omega)), \delta^0, 0)$.

LEMMA 3.5. *If $(M_t, \mathcal{F}_t, P)$ is a martingale and τ is an $\mathcal{F}_t$ -stopping time then there is a P -null set $N \in \mathcal{F}_\tau$ such that for $\omega \notin N$, $(M_t, \mathcal{F}_t, P_{\tau, \omega})$ is a martingale for $t \geq \tau_\omega$.*

Proof. The proof is comparable to that of Stroock and Varadhan [11, Thm. 1.2.10.] $\square$

COROLLARY 3.6. *If $P \in \mathcal{R}_{s, x}$ and τ is an $\mathcal{F}_t$ -stopping time, then there is a P -null set $N \in \mathcal{F}_\tau$ such that for $\omega \notin N$, $(M_t \phi, \mathcal{F}_t, P_{\tau, \omega} \circ \Theta_{\tau_\omega, \omega}^{-1})$ is a martingale for $t \geq \tau_\omega$ when $\phi \in C_b^2(\mathbb{R}^d)$.*

Proof. The proof is obvious from Lemmas 3.4 and 3.5. $\square$

The next two results are important for the rest of the paper. The first one states that a control rule remains a control rule for the problem starting at a later time from the point reached at that time. The second one says that if we take a control rule and at some later time switch to another control rule, then this concatenated object is still a control rule.

PROPOSITION 3.7 (closure under conditioning). *If $P \in \mathcal{R}_{s, x}$ and τ is a stopping time, $s \leq \tau \leq T$, then there exists a P -null set $N \in \mathcal{F}_\tau$ such that $\overset{\circ}{P}_\omega^\tau \in \mathcal{R}_{\tau_\omega, x_{\tau_\omega}}$ for $\omega \notin N$.*

Proof. Let $\{\phi_m\}$ be a dense subset of $C_0^\infty(\mathbb{R}^d)$. Then for each m , $(M_t \phi_m - M_{t \wedge \tau} \phi_m, \mathcal{F}_t, P)$ ($t \geq s$) is a martingale. For $\bar{\omega} \in \Omega$ define

$$\begin{aligned} \bar{M}_t^{\bar{\omega}} \phi_m &= \phi(x_t) - \phi(x_{t \wedge \tau_{\bar{\omega}}}) - \int_{t \wedge \tau_{\bar{\omega}}}^t \mathcal{L} \phi_m(\theta, x_\theta, \mu_\theta) d\theta \\ &\quad - \int_{t \wedge \tau_{\bar{\omega}}}^t \nabla_x \phi_m(x_\theta) \cdot g(\theta) dv_\theta^c - \sum_{t \wedge \tau_{\bar{\omega}} \leq \theta < t} [\phi_m(x_{\theta+}) - \phi_m(x_\theta)]. \end{aligned}$$

Then, by Lemma 3.5 there exists a P -null set $N_m \in \mathcal{F}_\tau$ such that $(\bar{M}_t^{\bar{\omega}} \phi_m, \mathcal{F}_t, P_{\tau, \bar{\omega}})$ is a martingale for $\bar{\omega} \notin N_m$. By Lemma 3.4 we know that $(\bar{M}_t^{\bar{\omega}} \phi_m, \mathcal{F}_t, P_{\tau, \bar{\omega}} \circ \Theta_{\tau_\omega, \bar{\omega}}^{-1})$ is a martingale for $t \geq \tau_{\bar{\omega}}$, $\bar{\omega} \notin N_m$. Certainly, $(\bar{M}_{t \wedge \tau_{\bar{\omega}}}^{\bar{\omega}} \phi_m, \mathcal{F}_t, \overset{\circ}{P}_{\bar{\omega}}^\tau)$ is a martingale, where $\bar{\omega}' = (\bar{x}_{\tau_{\bar{\omega}}}, \delta^0, 0)$. Therefore $(\bar{M}_{t \wedge \tau_{\bar{\omega}}}^{\bar{\omega}} \phi_m, \mathcal{F}_t, \overset{\circ}{P}_{\bar{\omega}}^\tau)$ is a martingale. It is obvious from the definition that

$$\overset{\circ}{P}_{\bar{\omega}}^\tau(\omega : x_r = \bar{x}_{\tau_{\bar{\omega}}}, \mu_r = \delta^0, v_r = 0, 0 \leq r \leq \tau_{\bar{\omega}}) = 1.$$

Let $N = \bigcup_m N_m$; then $P(N) = 0$. Through a limit procedure we can show that $\overset{\circ}{P}_\omega^\tau \in \mathcal{R}_{\tau_\omega, x_{\tau_\omega}}$ for $\omega \notin N$. $\square$

PROPOSITION 3.8 (closure under concatenation). *Let $P \in \mathcal{R}_{s,x}$ and τ be a stopping time such that $s \leq \tau \leq T$. If Q_ω is a transition probability such that $Q_\omega \in \mathcal{R}_{\tau(\omega),x(\tau(\omega))}$, then*

$$P \otimes_\tau Q \in \mathcal{R}_{s,x}.$$

Proof. It is obvious that we need only to show that for each $\phi \in C_b^2(\mathbb{R}^d)$, $(M_t \phi, \mathcal{F}_t, P \otimes_\tau Q)$ is a martingale after time s . From Remark 3.2 and the fact that $Q_\omega \in \mathcal{R}_{\tau(\omega),x(\tau(\omega))}$, it can be easily verified that $(M_t \phi, \mathcal{F}_t, \delta_\omega \otimes_\tau Q_\omega)$ is a martingale after time τ . By definition we know that $Q_\omega^\tau = \delta_\omega \otimes_\tau Q_\omega$ equals the regular conditional probability distribution of $P \otimes_\tau Q$ given $\mathcal{F}_\tau$. The proof of the proposition now follows Theorem 1.2.10 in Stroock and Varadhan [11]. $\square$

Set $Q_\omega = H(\tau(\omega), x(\tau(\omega)))$, where H is a measurable selector of $\mathcal{R}^o$ and, by definition, it is a τ -t.p. We denote

$$P \otimes_\tau H = P \otimes_\tau Q_\omega.$$

COROLLARY 3.9. *For P in $\mathcal{R}_{s,x}$ we have $P \otimes_\tau H \in \mathcal{R}_{s,x}$.*

3.2. The dynamic programming principle. With the preparations of §3.1, we can now establish the dynamic programming principle.

For a given probability $P \in \mathcal{R}_{s,x}$, define $\mathcal{R}_{s,x}^\tau(P)$ to be the set of probabilities in $\mathcal{R}_{s,x}$ which coincide with P up to time τ , i.e.,

$$\mathcal{R}_{s,x}^\tau(P) = \{P \otimes_\tau Q : Q_\omega \in \mathcal{R}_{\tau(\omega),x(\tau(\omega))} \text{ such that } Q_\omega \text{ is a } \tau\text{-t.p.}\} \subset \mathcal{R}_{s,x}.$$

We introduce the following notation: for a measurable function $\phi \in \mathcal{B}(\Sigma)$, let

$$\Gamma_s(t, \phi)(\omega) = \int_s^t f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t c(\theta) \cdot dv(\theta) + \phi(t, x_t),$$

where $\omega = (x, \mu, v) \in \Omega$.

THEOREM 3.10 (dynamic programming principle). (a) *If τ is an $\mathcal{F}_t$ -stopping time, $s \leq \tau \leq T$, and $P \in \mathcal{R}_{s,x}$, then*

$$(3.8) \quad \begin{aligned} & E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta + W(\tau, x_\tau) \right\} \\ &= \inf \left\{ \hat{P}(\Gamma_s) : \hat{P} \in \mathcal{R}_{s,x}^\tau(P) \right\}, \end{aligned}$$

where $\Gamma_s(\cdot)$ is defined by (2.5).

(b) *For $P \in \mathcal{R}_{s,x}$, $(\Gamma_s(t, W), \mathcal{F}_t, P)$ is a submartingale.*

(c) *If $s \leq \tau \leq T$, then*

$$(3.9) \quad W(s, x) = \inf \{ P\Gamma_s(\tau, W), P \in \mathcal{R}_{s,x} \}.$$

(d) *$(\Gamma_s(t, W), \mathcal{F}_t, P)$ is a martingale under P if and only if $P \in \mathcal{R}_{s,x}$ is optimal.*

Proof. (a) Recall that H denotes a measurable selector of $\mathcal{R}^o$; therefore, by (3.3) and Remark 3.2 the left-hand side (LHS) of (3.8) is

$$\begin{aligned} & E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta + H(\tau(\cdot), x_\tau(\cdot)) \Gamma_\tau \right\} \\ &= E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta + \delta_\cdot \otimes_\tau H(\Gamma_\tau) \right\} \\ &= P \otimes_\tau H(\Gamma_s), \end{aligned}$$

and by Proposition 3.8 we know that $P \otimes_\tau H \in \mathcal{R}_{s,x}^\tau(P)$. Hence, in (3.8) LHS $\geq$ RHS. On the other hand, for $\hat{P} = P \otimes_\tau Q \in \mathcal{R}_{s,x}^\tau(P)$,

$$\begin{aligned} \hat{P}(\Gamma_s) &= P \otimes_\tau Q(\Gamma_s) \\ &= E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta + Q(\Gamma_\tau) \right\} \\ &\geq E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta + W(\tau(\cdot), x_\tau(\cdot)) \right\} \\ &= \text{LHS of (3.8)}, \end{aligned}$$

so the proof of (a) is completed.

(b) For any $s \leq t < t+h$, we have

$$\begin{aligned} &E^P \{ \Gamma_s(t+h, W) - \Gamma_s(t, W) | \mathcal{F}_t \} \\ &= E^P \left\{ \int_s^{t+h} f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^{t+h} c(\theta) \cdot dv_\theta + W(t+h, x_{t+h}) \right. \\ &\quad \left. - \left[\int_s^t f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t c(\theta) \cdot dv_\theta + W(t, x_t) \right] \middle| \mathcal{F}_t \right\} \\ &= E^P \left\{ \int_t^{t+h} f(\theta, x_\theta, \mu_\theta) d\theta + \int_t^{t+h} c(\theta) \cdot dv_\theta + W(t+h, x_{t+h}) \middle| \mathcal{F}_t \right\} - W(t, x_t) \\ &= P_{t,\omega} \Gamma_t(t+h, W) - W(t, x_t) \\ &= \overset{\circ}{P}_\omega^t \Gamma_t(t+h, W) - W(t, x_t) \end{aligned}$$

by Lemma 3.3(2). Now we apply (3.8) and the fact that $\mathcal{R}_{t,x_t}^{t+h}(\overset{\circ}{P}_\omega^t) \subset \mathcal{R}_{t,x_t}$, i.e., Proposition 3.7, to get

$$\begin{aligned} &E^P \{ \Gamma_s(t+h, W) - \Gamma_s(t, W) | \mathcal{F}_t \} \\ &= \inf \left\{ \hat{P} \Gamma_t : \hat{P} \in \mathcal{R}_{t,x_t}^{t+h}(\overset{\circ}{P}_\omega^t) \right\} - W(t, x_t) \\ &\geq \inf \left\{ \hat{P} \Gamma_t : \hat{P} \in \mathcal{R}_{t,x_t} \right\} - W(t, x_t) \\ &= 0. \end{aligned}$$

Therefore $(\Gamma_s(t, W), \mathcal{F}_t, P)$ is a submartingale.

(c) Let $\nu(d\theta dx)$ be the distribution of (τ, x_τ) under P . Then

$$\begin{aligned} E^P W(\tau, x_\tau) &= \int W(\theta, x) \nu(d\theta dx) \\ &= \int J(\theta, H(\theta, x)) \nu(d\theta dx) \\ &= E^P J(\tau, H(\tau, x_\tau)) \\ &= J(\tau, P \otimes_\tau H). \end{aligned}$$

Note that

$$E^P \Gamma_s(\tau, W) = E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta \right\} + E^P W(\tau, x_\tau)$$

$$\begin{aligned}
&= E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta \right\} + J(\tau, P \otimes_\tau H) \\
&= J(s, P \otimes_\tau H) \\
&\geq \inf \{ J(s, Q), Q \in \mathcal{R}_{s,x} \} \\
&= W(s, x).
\end{aligned}$$

On the other hand, $E^P W(\tau, x_\tau) \leq E^P J(\tau, \overset{\circ}{P}_\omega^\tau) = E^P J(\tau, P_\omega^\tau) = J(\tau, P)$ and, therefore, by (b),

$$\begin{aligned}
(3.10) \quad W(s, x) &\leq E^P \Gamma_s(\tau, W) \\
&= E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta \right\} + E^P W(\tau, x_\tau) \\
&\leq E^P \left\{ \int_s^\tau f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^\tau c(\theta) \cdot dv_\theta \right\} + J(\tau, P) \\
&= J(s, P),
\end{aligned}$$

and thus (c) is proved if we take the infimum over $P \in \mathcal{R}_{s,x}$ on the right-hand side (RHS).

(d) If $\Gamma_s(t, W)$ is a P -martingale, then

$$W(s, x) = E^P \Gamma_s(s, W) = E^P \Gamma_s(T, W) = E^P \Gamma_s(T, 0) = E^P \Gamma_s = J(s, P),$$

because from our assumptions, $W(T, \cdot) = 0$. So P is optimal.

If we assume that $P \in \mathcal{R}_{s,x}$ is optimal, then by (3.10), Proposition 3.7, and Corollary 3.6,

$$(3.11) \quad W(s, x) \leq E^P \Gamma_s(t, W) = E^P \Gamma_s = W(s, x).$$

Therefore $(\Gamma_s(t, W), \mathcal{F}_t, P)$ is a submartingale with constant mean value, so it is indeed a martingale. $\square$

4. Continuity of the value function. In the rest of the paper we add the following assumptions:

- $c(\cdot)$ is Lipschitz continuous, $f(\cdot, \cdot, \cdot)$ is bounded, and $g = (g^{ij})$ is a constant $d \times k$ -matrix;
- $f(\cdot, \cdot, \cdot)$, $b(\cdot, \cdot, \cdot)$, $\sigma(\cdot, \cdot, \cdot)$ satisfy the following conditions:

$$\begin{aligned}
(4.1) \quad &|f(t, x, u) - f(s, y, u)| \leq C(|t - s| + \|x - y\|), \\
&\|b(t, x, u) - b(s, y, u)\| \leq C(|t - s| + \|x - y\|), \\
&\|\sigma(t, x, u) - \sigma(s, y, u)\| \leq C(|t - s| + \|x - y\|)
\end{aligned}$$

uniformly for $0 \leq s, t \leq T$, $x, y \in \mathbb{R}^d$, $u \in U$.

We will prove that under these conditions the value function $W(\cdot, \cdot)$ is uniformly continuous on Σ . In fact, there exists a constant $C \geq 0$ such that

$$|W(t, x) - W(s, y)| \leq C \left(|t - s|^{\frac{1}{2}} + \|x - y\| \right), \quad 0 \leq s, t \leq T, \quad x, y \in \mathbb{R}^d.$$

Note that the constancy of g is only required in the proof of Theorem 4.2.

THEOREM 4.1. *The value function $W(s, x)$ is uniformly Lipschitz continuous in the state variable x , i.e., there exists a constant $C > 0$ such that*

$$(4.2) \quad |W(s, x') - W(s, x)| \leq C \|x' - x\| \quad \forall 0 \leq t \leq T, \quad x, x' \in \mathbb{R}^d.$$

Proof. In the following, we use the same notation C to denote the constants, which may change from time to time. For any $0 \leq s \leq T$, $x, x' \in \mathbb{R}^d$,

$$W(s, x') - W(s, x) \leq \sup_{P \in \mathcal{R}_{s,x}} (E^Q \Gamma_s - E^P \Gamma_s)$$

for each $Q \in \mathcal{R}_{s,x'}$, where Γ is the cost function defined by (2.5).

Take an arbitrary $P \in \mathcal{R}_{s,x}$. By the definition of control rules, there exists a standard extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$ of $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, i.e., there exists another probability space $(\Omega', \mathcal{F}', \mathcal{F}'_t, P')$ such that $\tilde{\Omega} = \Omega \times \Omega'$, $\tilde{\mathcal{F}} = \mathcal{F} \times \mathcal{F}'$, $\tilde{\mathcal{F}}_t = \mathcal{F}_t \times \mathcal{F}'_t$, and $\tilde{P} = P \times P'$. We can extend the processes x, μ, v to $\tilde{\Omega}$ by the following: for $\tilde{\omega} = (\omega, \omega') \in \tilde{\Omega}$,

$$x.(\tilde{\omega}) = x.(\omega), \mu.(\tilde{\omega}) = \mu.(\omega), v.(\tilde{\omega}) = v.(\omega).$$

On $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$ there exists a standard d -dimensional Brownian motion B such that for $s \leq t \leq T$,

$$(4.3) \quad x_t = x + \int_s^t b(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, \mu_\theta) dB_\theta + g v_t \quad \text{a.s.}$$

Consider the same equation (4.3) with the initial state x' , i.e.,

$$(4.4) \quad y_t = x' + \int_s^t b(\theta, y_\theta, \mu_\theta) d\theta + \int_s^t \sigma(\theta, y_\theta, \mu_\theta) dB_\theta + g v_t$$

on the stochastic basis $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$. The strong solution for (4.4) exists from the assumptions on $b(\cdot, \cdot, \cdot)$ and $\sigma(\cdot, \cdot, \cdot)$, and so $\alpha \equiv (\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P}, y_t, \mu_t, v_t, s, x') \in \tilde{\Lambda}_{s,x'}$. Therefore, there exists a control rule $Q \in \mathcal{R}_{s,x'}$ such that

$$(4.5) \quad J(\alpha) = J(s, Q) = E^Q \Gamma_s.$$

By definition,

$$\begin{aligned} E^P \Gamma_s &= E^P \left\{ \int_s^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta \right\} \\ &= E^{\tilde{P}} \left\{ \int_s^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta \right\}, \end{aligned}$$

and, by (4.5),

$$E^Q \Gamma_s = E^{\tilde{P}} \left\{ \int_s^T f(\theta, y_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta \right\};$$

therefore, from the Lipschitz continuity of f we have

$$\begin{aligned} E^Q \Gamma_s - E^P \Gamma_s &\leq E^{\tilde{P}} \left\{ \int_s^T |f(\theta, y_\theta, \mu_\theta) - f(\theta, x_\theta, \mu_\theta)| d\theta \right\} \\ &\leq C E^{\tilde{P}} \int_s^T \|y_\theta - x_\theta\| d\theta \\ &\leq C \left(\int_s^T E^{\tilde{P}} \|y_\theta - x_\theta\|^2 d\theta \right)^{\frac{1}{2}}. \end{aligned}$$

Now, from equations (4.3), (4.4), and the Lipschitz continuity conditions on b , σ , we have for $\theta \geq s$,

$$\begin{aligned} E^{\tilde{P}} \|y_\theta - x_\theta\|^2 &\leq C \|x' - x\|^2 + CE^{\tilde{P}} \left(\int_s^\theta \|b(h, y_h, \mu_h) - b(h, x_h, \mu_h)\| dh \right)^2 \\ &\quad + CE^{\tilde{P}} \sup_{\theta' \leq \theta} \left\| \int_s^{\theta'} (\sigma(h, y_h, \mu_h) - \sigma(h, x_h, \mu_h)) dB_h \right\|^2 \\ &\leq C \|x' - x\|^2 + CE^{\tilde{P}} \int_s^\theta \|b(h, y_h, \mu_h) - b(h, x_h, \mu_h)\|^2 dh \\ &\quad + CE^{\tilde{P}} \int_s^\theta \|\sigma(h, y_h, \mu_h) - \sigma(h, x_h, \mu_h)\|^2 dh \\ &\leq C \left(\|x' - x\|^2 + \int_s^\theta E^{\tilde{P}} \|y_h - x_h\|^2 dh \right). \end{aligned}$$

We have used the Burkholder–Gundy inequality to get the second inequality. By Gronwall’s inequality,

$$E^{\tilde{P}} \|y_\theta - x_\theta\|^2 \leq C \|x' - x\|^2 e^{C(\theta-s)} \leq C \|x' - x\|^2.$$

Hence we have

$$E^Q \Gamma_s - E^P \Gamma_s \leq C \|x' - x\|$$

and, therefore, $W(s, x,) - W(s, x) \leq C \|x' - x\|$.

The proof of the theorem is thus complete since $x, x' \in \mathbb{R}^d$ are arbitrary. $\square$

Next we consider the continuity of the value function in the time variable t .

THEOREM 4.2. *The value function $W(t, x)$ is uniformly continuous in the time variable t . In fact, there exists a constant $C > 0$ such that*

$$(4.6) \quad |W(s, x) - W(s', x)| \leq C |s - s'|^{\frac{1}{2}}$$

for all $0 \leq s, s' \leq T, x \in \mathbb{R}^d$.

Proof. As in the proof of Theorem 4.1, we use the same notation C to denote the constants. First we assume $s \leq s'$, so that

$$(4.7) \quad W(s', x) - W(s, x) \leq \sup_{P \in \mathcal{R}_{s,x}} (E^Q \Gamma_{s'} - E^P \Gamma_s)$$

for each $Q \in \mathcal{R}_{s',x}$. From the strict positivity of $c(\cdot)$, we may actually take the supremum in (4.7) over a subset $\mathcal{R}_{s,x}(\lambda)$ for some $\lambda > 0$, where

$$(4.8) \quad \mathcal{R}_{s,x}(\lambda) = \{P \in \mathcal{R}_{s,x} : E^P \|v(T)\| \leq \lambda\}.$$

Now, for $P \in \mathcal{R}_{s,x}(\lambda)$, as in the proof Theorem 4.1, there exists a standard extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$ of $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ such that $x(\cdot)$ is a solution of

$$(4.9) \quad x_t = x + \int_s^t b(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, \mu_\theta) dB_\theta + g v_t$$

on the stochastic basis $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$, where B is a d -dimensional Brownian motion. Define $\tilde{\mathcal{F}}_t = \tilde{\mathcal{F}}_{t-s'+s}$, $\hat{\mu}_t = \mu_{t-s'+s}$, $\hat{v}_t = v_{t-s'+s}$, and $\hat{B}_t = B_{t-s'+s}$ for $t \geq s'$. Consider the following stochastic differential equation:

$$(4.10) \quad y_t = x + \int_{s'}^t b(\theta, y_\theta, \hat{\mu}_\theta) d\theta + \int_{s'}^t \sigma(\theta, y_\theta, \hat{\mu}_\theta) d\hat{B}_\theta + g\hat{v}_t, \quad t \geq s'$$

on the stochastic basis $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P})$. We know that under assumption (4.1) there exists a unique strong solution y , and by definition, $\alpha = (\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathcal{F}}_t, \tilde{P}, y_t, \hat{\mu}_t, \hat{v}_t, s', x) \in \tilde{\Lambda}_{s',x}$. Therefore, there exists a control rule $Q \in \mathcal{R}_{s',x}$ such that

$$(4.11) \quad J(\alpha) = J(s', Q) = E^Q \Gamma_{s'}.$$

Recall that Γ_s is defined by (2.5). Thus, by definition,

$$E^P \Gamma_s = E^{\tilde{P}} \left\{ \int_s^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^T c(\theta) \cdot dv_\theta \right\},$$

and by (4.11),

$$\begin{aligned} E^Q \Gamma_{s'} &= E^{\tilde{P}} \left\{ \int_{s'}^T f(\theta, y_\theta, \hat{\mu}_\theta) d\theta + \int_{s'}^T c(\theta) \cdot d\hat{v}_\theta \right\} \\ &= E^{\tilde{P}} \left\{ \int_s^{T-s'+s} f(\theta + s' - s, y_{\theta+s'-s}, \mu_\theta) d\theta + \int_s^{T-s'+s} c(\theta + s' - s) \cdot dv_\theta \right\}. \end{aligned}$$

Therefore, noticing that f is bounded below by a constant $-K$ ($K \geq 0$) and $P \in \mathcal{R}_{s,x}(\lambda)$, we get

$$\begin{aligned} E^Q \Gamma_{s'} - E^P \Gamma_s &\leq E^{\tilde{P}} \left\{ \int_s^{T-(s'-s)} |f(\theta, x_\theta, \mu_\theta) - f(\theta + s' - s, y_{\theta+s'-s}, \mu_\theta)| d\theta \right. \\ &\quad \left. + \int_s^{T-(s'-s)} \|c(\theta) - c(\theta + s' - s)\| d\|v_\theta\| + K|s' - s| \right\} \\ &\leq E^{\tilde{P}} \left\{ \int_s^{T-(s'-s)} |f(\theta, x_\theta, \mu_\theta) - f(\theta + s' - s, y_{\theta+s'-s}, \mu_\theta)| d\theta \right\} \\ &\quad + C|s' - s| (E^P \|v(T)\| + 1). \end{aligned}$$

From the Lipschitz continuity of the function f , we have

$$|f(\theta, x_\theta, \mu_\theta) - f(\theta + s' - s, y_{\theta+s'-s}, \mu_\theta)| \leq C(|s' - s| + \|x_\theta - y_{\theta+s'-s}\|).$$

Therefore

$$(4.12) \quad E^Q \Gamma_{s'} - E^P \Gamma_s \leq C \left(|s' - s| + E^{\tilde{P}} \int_s^{T-(s'-s)} \|x_\theta - y_{\theta+s'-s}\| d\theta \right).$$

By (4.10) we have for $\theta \geq s$,

$$(4.13) \quad y_{\theta+s'-s} = x + \int_{s'}^{\theta+s'-s} b(h, y_h, \hat{\mu}_h) dh$$

$$\begin{aligned}
& + \int_{s'}^{\theta+s'-s} \sigma(h, y_h, \hat{\mu}_h) d\hat{B}_u + g\hat{v}_{\theta+s'-s} \\
& = x + \int_s^\theta b(h+s'-s, y_{h+s'-s}, \mu_h) dh \\
& \quad + \int_s^\theta \sigma(h+s'-s, y_{h+s'-s}, \mu_h) dB_h + gv_\theta.
\end{aligned}$$

So from (4.9), (4.13), and the Burkholder–Davis–Gundy inequality, we have for $\theta \geq s$,

$$\begin{aligned}
& E^{\tilde{P}} \|x_\theta - y_{\theta+s'-s}\|^2 \\
& \leq 2E \left(\int_s^\theta \|b(h, x_h, \mu_h) - b(h+s'-s, y_{h+s'-s}, \mu_h)\| dh \right)^2 \\
& \quad + 2E^{\tilde{P}} \sup_{\theta' \leq \theta} \left\| \int_s^{\theta'} (\sigma(h, x_h, \mu_h) - \sigma(h+s'-s, y_{h+s'-s}, \mu_h)) dB_h \right\|^2 \\
& \leq 2TE^{\tilde{P}} \int_s^\theta \|b(h, x_h, \mu_h) - b(h+s'-s, y_{h+s'-s}, \mu_h)\|^2 dh \\
& \quad + 2E^{\tilde{P}} \int_s^\theta \|\sigma(h, x_h, \mu_h) - \sigma(h+s'-s, y_{h+s'-s}, \mu_h)\|^2 dh \\
& \leq CE^{\tilde{P}} \int_s^\theta (|s'-s|^2 + \|x_h - y_{h+s'-s}\|^2) dh \\
& \leq C \left(|s'-s|^2 + \int_s^\theta E^{\tilde{P}} \|x_h - y_{h+s'-s}\|^2 dh \right).
\end{aligned}$$

Gronwall's inequality implies

$$E^{\tilde{P}} \|x_\theta - y_{\theta+s'-s}\|^2 \leq C|s'-s|^2 e^{C(\theta-s)} \leq C|s'-s|^2.$$

Hence, from (4.12) we have

$$E^Q \Gamma_{s'} - E^P \Gamma_s \leq C|s'-s|$$

and, therefore, $W(s', x) - W(s, x) \leq C|s'-s|$ for $s' > s$.

Now we assume $s' < s$. By the dynamic programming principle (cf. Theorem 3.10),

$$W(s', x) = \inf_{P \in \mathcal{R}_{s', x}} E^P \left\{ \int_{s'}^s f(\theta, x_\theta, \mu_\theta) d\theta + \int_{s'}^s c(\theta) \cdot dv_\theta + W(s, x_s) \right\}.$$

Take $P^0 \in \mathcal{R}_{s', x}$ such that $P^0(\mu_\theta = \delta_{\{u^0\}}, v_\theta = 0, 0 \leq \theta \leq T) = 1$ for some arbitrary but fixed $u^0 \in U$. Then

$$\begin{aligned}
W(s', x) & \leq E^{P^0} \left\{ \int_{s'}^s f(\theta, x_\theta, u^0) d\theta + W(s, x_s) \right\} \\
& \leq C|s'-s| + E^{P^0} W(s, x_s)
\end{aligned}$$

by the boundedness of f , and by Theorem 4.1 there exists a constant C such that

$$W(s, x_s) \leq W(s, x) + C\|x_s - x\|.$$

Hence

$$(4.14) \quad \begin{aligned} W(s', x) - W(s, x) &\leq C(|s' - s| + E^{P^0} \|x_s - x\|) \\ &\leq C \left(|s' - s| + (E^{P^0} \|x_s - x\|^2)^{\frac{1}{2}} \right). \end{aligned}$$

From the definition of control rules, we know that under P^0 ,

$$(4.15) \quad x_s - x = \int_{s'}^s b(\theta, x_\theta, u^0) d\theta + M_s,$$

where M is a continuous square integrable martingale with

$$\langle M \rangle_s = \int_{s'}^s \text{tr}(a(\theta, x_\theta, u^0)) d\theta.$$

Therefore, by (4.15) and the Burkholder–Davis–Gundy inequality, we have

$$(4.16) \quad E^{P^0} \|x_s - x\|^2 \leq C(|s' - s|^2 + |s' - s|).$$

Combining (4.14) and (4.16) we have

$$W(s', x) - W(s, x) \leq C |s' - s|^{\frac{1}{2}}.$$

The theorem is thus proved. $\square$

Remark 4.3. We have assumed that the function f is bounded. It is easy to see from the proof of Theorem 4.2 that without this condition the constant C will depend on x .

Combining Theorems 4.1 and 4.2 we can state the main result of this section.

THEOREM 4.4. *The value function W is uniformly continuous on Σ . Moreover, there exists a constant $C \geq 0$ such that*

$$|W(t, x) - W(s, y)| \leq C \left(|t - s|^{\frac{1}{2}} + \|x - y\| \right), \quad 0 \leq s, t \leq T, \quad x, y \in \mathbb{R}^d.$$

5. The dynamic programming equation. Before we derive the dynamic programming equation heuristically, we prove a result which shows that there exists a set such that the optimal state process is continuous when it is in this set.

THEOREM 5.1. (a) *Assume $(t, x) \in \Sigma$; then*

$$(5.1) \quad W(t, x) \leq W(t, x + gh) + c(t) \cdot h$$

for each $h \in \mathbb{R}_+^k$. Moreover, if equality holds for some $h = (h^i) \in \mathbb{R}^k$, then the same equality holds when we replace h by $\bar{h}$ with $\bar{h} = (\bar{h}^i) \in \mathbb{R}_+^k$, $\bar{h}^i \leq h^i$ ($1 \leq i \leq k$).

(b) *Define, for $0 \leq t \leq T$,*

$$A_t \equiv \{x : W(t, x) < W(t, x + gh) + c(t) \cdot h, \forall h \in \mathbb{R}_+^k, h \neq 0\}.$$

Then the optimal state process x_t is continuous when it is in A_t . To be precise, we have

$$(5.2) \quad P(\Delta x_t \neq 0, x_t \in A_t) = 0, \quad s \leq t \leq T$$

for every $P \in \mathcal{R}_{s,x}^o$, $(s, x) \in \Sigma$.

Proof. (a) If (5.1) fails for some $h \in \mathbb{R}_+^k$, then

$$(5.3) \quad W(t, x) > W(t, x + gh) + c(t) \cdot h.$$

Take $P \in \mathcal{R}_{t, x+gh}^o$. We define $\Theta : \Omega \rightarrow \Omega$ by

$$\Theta(\omega)_s = \begin{cases} (x_s - gh, \mu_s, 0), & 0 \leq s \leq t, \\ (x_s, \mu_s, v_s + h), & t < s \leq T \end{cases}$$

for $\omega = (x, \mu, v)$, and let $\tilde{P} = P \circ \Theta^{-1}(\cdot)$. As in the proof of Lemma 3.4, we can show that $\tilde{P} \in \mathcal{R}_{t, x}$. From the definition of $\tilde{P}$ we have

$$\begin{aligned} J(t, \tilde{P}) &= E^{\tilde{P}} \left\{ \int_t^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_t^T c(\theta) \cdot dv_\theta \right\} \\ &= E^P \left\{ \int_t^T f(\theta, x_\theta, \mu_\theta) d\theta + \int_t^T c(\theta) \cdot dv_\theta + c(t) \cdot h \right\} \\ &= J(t, P) + c(t) \cdot h \\ &= W(t, x + gh) + c(t) \cdot h. \end{aligned}$$

Therefore, $J(t, \tilde{P}) < W(t, x)$ from (5.3), a contradiction.

Next, if (5.1) holds as an equality for h , then for $\bar{h}, \bar{h}^i \leq h^i, 1 \leq i \leq k$,

$$\begin{aligned} W(t, x) - c(t) \cdot h &= W(t, x + gh) \\ &\geq W(t, x + g\bar{h}) - c(t) \cdot (h - \bar{h}) \\ &\geq W(t, x) - c(t) \cdot \bar{h} - c(t) \cdot (h - \bar{h}) \\ &= W(t, x) - c(t) \cdot h. \end{aligned}$$

Therefore

$$W(t, x) = W(t, x + g\bar{h}) + c(t) \cdot \bar{h}.$$

(b) For $P \in \mathcal{R}_{s, x}$, we know that P -a.s.,

$$\begin{aligned} x_t &= x + \int_s^t b(\theta, x_\theta, \mu_\theta) d\theta + g v_t \\ &\quad + \text{a continuous local martingale.} \end{aligned}$$

Thus $\Delta x_t = g \Delta v_t$, and $x_{t+} = x_t + \Delta x_t = x_t + g \Delta v_t$. Since $W(\cdot, \cdot)$ is continuous on Σ , for $t \leq s \leq T$,

$$(5.4) \quad W(t, x_{t+}) = \lim_{t' \downarrow t} W(t', x_{t'}).$$

Assume $P \in \mathcal{R}_{s, x}^o$, then by the dynamic programming principle (cf. Theorem 3.10) we know that $(\Gamma_s(t, W), \mathcal{F}_t, P)$ is a martingale. Hence, for $t' > t$,

$$(5.5) \quad W(s, x) = E^P \left\{ \int_s^{t'} f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^{t'} c(\theta) \cdot dv_\theta + W(t', x_{t'}) \right\}.$$

Let $t' \downarrow t$; note that from our assumption we know that $c(\cdot)$ is continuous on $[0, T]$. Therefore

$$\begin{aligned} \int_s^{t'} f(\theta, x_\theta, \mu_\theta) d\theta &\rightarrow \int_s^t f(\theta, x_\theta, \mu_\theta) d\theta, \\ \int_s^{t'} c(\theta) \cdot dv_\theta &\rightarrow \int_s^t c(\theta) \cdot dv_\theta + c(t) \cdot \Delta v_t. \end{aligned}$$

Hence if (5.2) fails, then (5.4) and (5.5) imply

$$\begin{aligned} W(s, x) &= E^P \left\{ \int_s^t f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t c(\theta) \cdot dv_\theta + c(t) \cdot \Delta v_t + W(t, x_t + g\Delta v_t) \right\} \\ &> E^P \left\{ \int_s^t f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t c(\theta) \cdot dv_\theta + W(t, x_t) \right\} \\ &= E^P \Gamma_s(t, W), \end{aligned}$$

which contradicts the fact that $P \in \mathcal{R}_{s,x}^o$. The inequality follows from the facts that

$$W(t, x_t) \leq W(t, x_t + g\Delta x_t) + c(t) \cdot \Delta v_t$$

and the strict inequality holds if $x_t \in A_t$ and $\Delta x_t \neq 0$. $\square$

Remark 5.2. From Theorem 5.1 (a), we can see that if the value function $W \in C^{1,2}(\Sigma)$, then

$$(g^* \nabla_x W(t, x))^i + c^i(t) \geq 0, \quad i = 1, 2, \dots, k,$$

where $*$ means transpose and $(\cdot)^i$ denotes the i th coordinate of the point in $\mathbb{R}^k$. For $x \notin A_t$, there exists $h = (h^i) \in \mathbb{R}_+^k$ such that for $\bar{h} = (\bar{h}^i) \in \mathbb{R}_+^k$, $\bar{h}^i \leq h^i$, $1 \leq i \leq k$,

$$W(t, x) = W(t, x + g\bar{h}) + c(t) \cdot \bar{h}.$$

Therefore we have

$$(g^* \nabla_x W(t, x))^i + c^i(t) = 0$$

for those i such that $h^i > 0$.

5.1. Heuristic derivation of the dynamic programming equation. Recall Ito's formula. For $\phi \in C^{1,2}(\Sigma)$, $(s, x) \in \Sigma$, $t \geq s$,

$$\begin{aligned} (5.6) \quad \phi(t, x_t) &= \phi(s, x) + \int_s^t \left(\frac{\partial}{\partial t} + \mathcal{L} \right) \phi(\theta, x_\theta, \mu_\theta) d\theta \\ &\quad + \int_s^t \nabla_x \phi(\theta, x_\theta) \cdot a(\theta, x_\theta, \mu_\theta) dB_\theta + \int_s^t \nabla_x \phi(\theta, x_\theta) \cdot g dv_\theta \\ &\quad + \sum_{s \leq \theta < t} [\phi(\theta, x_{\theta+}) - \phi(\theta, x_\theta) - \nabla_x \phi(\theta, x_\theta) \cdot \Delta x_\theta]. \end{aligned}$$

Let

$$\tilde{\mathcal{L}} \equiv \frac{\partial}{\partial t} + \mathcal{L} = \frac{\partial}{\partial t} + \frac{1}{2} \sum_{i,j} a_{ij} \frac{\partial^2}{\partial x^i \partial x^j} + \sum_i b_i \frac{\partial}{\partial x^i};$$

then for $P \in \mathcal{R}_{s,x}$, (5.6) may be written as

$$E^P \phi(t, x_t) = \phi(s, x) + E^P \left\{ \int_s^t \tilde{\mathcal{L}}\phi(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t \nabla_x \phi(\theta, x_\theta) \cdot g dv_\theta \right. \\ \left. + \sum_{s \leq \theta < t} [\phi(\theta, x_{\theta+}) - \phi(\theta, x_\theta) - \nabla_x \phi(\theta, x_\theta) \cdot \Delta x_\theta] \right\}.$$

By the dynamic programming principle (cf. Theorem 3.10),

$$W(s, x) = \inf_{P \in \mathcal{R}_{s,x}} E^P \left\{ \int_s^t f(\theta, x_\theta, \mu_\theta) d\theta + \int_s^t c(\theta) \cdot dv_\theta + W(t, x_t) \right\},$$

so if we assume $W \in C^{1,2}(\Sigma)$, then

$$(5.7) \quad 0 = \inf_{P \in \mathcal{R}_{s,x}} E^P \left\{ \int_s^t (\tilde{\mathcal{L}}W + f)(\theta, x_\theta, \mu_\theta) d\theta \right. \\ \left. + \int_s^t (c(\theta) + g^* \nabla_x W(\theta, x_\theta)) \cdot dv_\theta \right. \\ \left. + \sum_{s \leq \theta < t} [W(\theta, x_{\theta+}) - W(\theta, x_\theta) - \nabla_x W(\theta, x_\theta) \cdot \Delta x_\theta] \right\}.$$

If we take the infimum over all those $P \in \mathcal{R}_{s,x}$ such that $P(v_t = 0, 0 \leq t \leq T) = 1$, then from (5.7) we get

$$\inf_{P \in \mathcal{R}_{s,x}} E^P \left\{ \int_s^t (\tilde{\mathcal{L}}W + f)(\theta, x_\theta, \mu_\theta) d\theta \right\} \geq 0.$$

Let $t \downarrow s$; we have

$$\inf_{P \in \mathcal{R}_{s,x}} E^P (\tilde{\mathcal{L}}W + f)(s, x_s, \mu_s) \geq 0.$$

Moreover, from Remark 5.2 we can conclude that on Σ ,

$$(5.8) \quad (g^* \nabla_x W)^i + c^i \geq 0, \quad i = 1, 2, \dots, k.$$

Therefore we can expect that W formally satisfies the following *variational inequality*, or *Hamilton–Jacobi–Bellman equation*,

$$(5.9) \quad \min \left\{ \inf_{u \in U} (\tilde{\mathcal{L}}W + f)(t, x, u), (g^* \nabla_x W(t, x))^i + c^i(t), i = 1, 2, \dots, k \right\} = 0$$

on Σ . For simplicity of notation, we write (5.9) as

$$\min \left\{ \inf_{u \in U} (\tilde{\mathcal{L}}W + f), g^* \nabla_x W + c \right\} = 0.$$

5.2. Viscosity solution. As is well known in classical control problems, the value function is a solution to the corresponding Hamilton–Jacobi–Bellman equation when it has sufficient regularity (cf. Fleming and Rishel [4], Krylov [9]). If it is only known that the value function is continuous, then, as observed by Lions [10], the value function is a solution to the Hamilton–Jacobi–Bellman equation in the *viscosity* sense, which we will define as follows.

DEFINITION 5.3. A function ψ is a viscosity solution of (5.9) if $\psi \in C(\Sigma)$ and, for every $\phi \in C^{1,2}(\Sigma)$,

(1) for each local maximum point (t_0, x_0) of $\psi - \phi$ in the interior of Σ , we have

$$(5.10) \quad \min \left\{ \inf_{u \in U} (\tilde{\mathcal{L}}\phi + f), g^* \nabla_x \phi + c \right\} \geq 0$$

at (t_0, x_0) , i.e., ψ is a subsolution;

(2) for each local minimum point (t_0, x_0) of $\psi - \phi$ in the interior of Σ , we have

$$(5.11) \quad \min \left\{ \inf_{u \in U} (\tilde{\mathcal{L}}\phi + f), g^* \nabla_x \phi + c \right\} \leq 0$$

at (t_0, x_0) , i.e., ψ is a supersolution.

For an introduction to the viscosity solution and its applications to stochastic optimal control problems, see Fleming and Soner [5]. For an extensive bibliography on viscosity solutions, see Crandall, Ishii, and Lions [2].

THEOREM 5.4. *The value function $W(\cdot, \cdot)$ is a viscosity solution of (5.9).*

Proof. By Theorem 4.4 we know $W \in C(\Sigma)$. We first show that W is a subsolution. For $\phi \in C^{1,2}(\Sigma)$, if $(t_0, x_0) \in \text{int}(\Sigma)$ is a local maximum point of $W - \phi$, then there is a neighborhood $O^1(t_0, x_0)$ of (t_0, x_0) in Σ such that

$$W(t, x) - \phi(t, x) \leq W(t_0, x_0) - \phi(t_0, x_0), \quad (t, x) \in \bar{O}^1(t_0, x_0),$$

or

$$(5.12) \quad W(t, x) - W(t_0, x_0) \leq \phi(t, x) - \phi(t_0, x_0), \quad (t, x) \in \bar{O}^1(t_0, x_0),$$

where $\bar{O}^i(t_0, x_0)$ denotes the closure of $O^i(t_0, x_0)$. If (5.10) fails, then one of the following will be true:

$$(5.13) \quad \inf_{u \in U} (\tilde{\mathcal{L}}\phi + f)(t_0, x_0, u) < 0,$$

$$(5.14) \quad (g^* \nabla_x \phi(t_0, x_0))^i + c^i(t_0) < 0 \text{ for some } 1 \leq i \leq k.$$

If (5.13) is true, then from assumption (4.1) and $\phi \in C^{1,2}(\Sigma)$, we can find $u^0 \in U$ and a neighborhood $O^2(t_0, x_0)$ of (t_0, x_0) such that

$$(\tilde{\mathcal{L}}\phi + f)(t, x, u^0) < 0$$

for $(t, x) \in O^2(t_0, x_0)$. Take $P \in \mathcal{R}_{t_0, x_0}$ such that

$$(5.15) \quad P(\mu_r = \delta_{\{u^0\}}, v_r = 0, 0 \leq r \leq T) = 1.$$

The existence of such a $P \in \mathcal{R}_{t_0, x_0}$ is obvious. Define

$$\tau = \inf \{ t > t_0, (t, x_t) \notin O(t_0, x_0) \},$$

where $O(t_0, x_0) = O^1(t_0, x_0) \cap O^2(t_0, x_0)$. Since the state process x is continuous a.s.(P), we can see immediately that

$$P(\tau > t_0) = 1,$$

and for $t_0 \leq \theta < \tau$,

$$(\tilde{\mathcal{L}}\phi + f)(\theta, x_\theta, \mu_\theta) < 0 \quad \text{a.s.}(P).$$

Therefore

$$E^P \int_{t_0}^{\tau} (\tilde{\mathcal{L}}\phi + f)(\theta, x_{\theta}, \mu_{\theta}) d\theta < 0.$$

By the definition of control rules,

$$\phi(\tau, x_{\tau}) = \phi(t_0, x_0) + \int_{t_0}^{\tau} \tilde{\mathcal{L}}\phi(\theta, x_{\theta}, \mu_{\theta}) d\theta + M_{\tau}\phi \text{ a.s.}(P),$$

where $M\phi \in \mathcal{M}_2^c$, i.e., a continuous square integrable martingale with respect to P . Note that Theorem 4.2.1 of Stroock and Varadhan [11] allows us to replace $\mathcal{L}$ by $\tilde{\mathcal{L}}$ in (2.4), at least when $v = 0$. Hence

$$E^P \phi(\tau, x_{\tau}) - \phi(t_0, x_0) = E^P \int_{t_0}^{\tau} \tilde{\mathcal{L}}\phi(\theta, x_{\theta}, \mu_{\theta}) d\theta.$$

Noting that $(\tau, x_{\tau}) \in \bar{O}(t_0, x_0)$, we have

$$\begin{aligned} E^P W(\tau, x_{\tau}) - W(t_0, x_0) &\leq E^P \phi(\tau, x_{\tau}) - \phi(t_0, x_0) \\ &= E^P \int_{t_0}^{\tau} \tilde{\mathcal{L}}\phi(\theta, x_{\theta}, \mu_{\theta}) d\theta \\ &< -E^P \int_{t_0}^{\tau} f(\theta, x_{\theta}, \mu_{\theta}) d\theta, \end{aligned}$$

which, by (5.15) can be rewritten as

$$\begin{aligned} W(t_0, x_0) &> E^P \left\{ \int_{t_0}^{\tau} f(\theta, x_{\theta}, \mu_{\theta}) d\theta + W(\tau, x_{\tau}) \right\} \\ &= E^P \left\{ \int_{t_0}^{\tau} f(\theta, x_{\theta}, \mu_{\theta}) d\theta + \int_{t_0}^{\tau} c(\theta) \cdot dv_{\theta} + W(\tau, x_{\tau}) \right\}. \end{aligned}$$

This contradicts the dynamic programming principle (cf. (3.9)).

Next, if (5.14) holds at (t_0, x_0) for some i , then we can take $h^i > 0$ small enough such that

$$\phi(t_0, x_0 + g^i h^i) - \phi(t_0, x_0) < -c^i(t_0) h^i,$$

where g^i denotes the i th column of the $d \times k$ matrix g . Therefore, by (5.12) we have

$$W(t_0, x_0 + g^i h^i) - W(t_0, x_0) < -c^i(t_0) h^i$$

and, therefore,

$$W(t_0, x_0) > W(t_0, x_0 + gh) + c(t_0) \cdot h,$$

where $h = (0, \dots, h^i, \dots, 0)$. This is a contradiction of Theorem 5.1, and thus we have shown that W is a subsolution of (5.9)

Now we show that W is also a supersolution of (5.9). If $\phi \in C^{1,2}(\Sigma)$ such that $W - \phi$ has a local minimum point at $(t_0, x_0) \in \text{int}(\Sigma)$, then there exists a neighborhood $O^1(t_0, x_0)$ of (t_0, x_0) satisfying

$$(5.16) \quad W(t, x) - W(t_0, x_0) \geq \phi(t, x) - \phi(t_0, x_0), \quad (t, x) \in \bar{O}^1(t_0, x_0).$$

If (5.11) fails, then

$$\inf_{u \in U} (\tilde{\mathcal{L}}\phi + f) > 0, \quad (g^*\nabla\phi)^i + c^i > 0$$

at (t_0, x_0) for $i = 1, \dots, k$. From assumption (4.1) and the fact that $\phi \in C^{1,2}(\Sigma)$, we can find a neighborhood $O^2(t_0, x_0)$ of (t_0, x_0) such that for some $\varepsilon > 0$,

$$\inf_{u \in U} (\tilde{\mathcal{L}}\phi + f) > \varepsilon, \quad (g^*\nabla\phi)^i + c^i > \varepsilon$$

on $\bar{O}^2(t_0, x_0)$ for $i = 1, \dots, k$. Let $O(t_0, x_0) = O^1(t_0, x_0) \cap O^2(t_0, x_0)$; then for $(t, x) \in O(t_0, x_0)$, we have for small $h \in \mathbb{R}_+^k$, $h \neq 0$,

$$\phi(t, x + gh) - \phi(t, x) > -c(t) \cdot h.$$

Therefore, by (5.16),

$$W(t, x + gh) - W(t, x) > -c(t) \cdot h$$

or $x \in A_t$. Hence for $P \in \mathcal{R}_{t_0, x_0}^o$,

$$(5.17) \quad P(x_{t_0+} = x_{t_0}) = 1$$

by Theorem 5.1. Define

$$\tau = \inf\{t \geq t_0, \quad (t, x_t) \notin O(t_0, x_0)\};$$

then from (5.17) we see that $P(\tau > t_0) = 1$ for $P \in \mathcal{R}_{t_0, x_0}^o$, and it can be seen that

$$(t, x_t) \in \bar{O}(t_0, x_0), \quad x_t \in A_t, \quad t_0 \leq t \leq \tau.$$

Therefore we have

$$E^P \left\{ \int_{t_0}^{\tau} (\tilde{\mathcal{L}}\phi + f)(\theta, x_\theta, \mu_\theta) d\theta + \int_{t_0}^{\tau} (g^*\nabla_x \phi(\theta, x_\theta) + c_\theta) \cdot dv_\theta \right\} \geq \varepsilon E^P(\tau - t_0).$$

Applying Ito's formula and noting that the state process x is continuous a.s. (P) when $t_0 \leq t \leq \tau$, we have

$$E^P \phi(\tau, x_\tau) = \phi(t_0, x_0) + E^P \left\{ \int_{t_0}^{\tau} \tilde{\mathcal{L}}\phi(\theta, x_\theta, \mu_\theta) d\theta + \int_{t_0}^{\tau} \nabla_x \phi(\theta, x_\theta) \cdot g dv_\theta \right\},$$

which may be rewritten as

$$E^P [\phi(\tau, x_\tau) - \phi(t_0, x_0)] \geq E^P \left\{ \int_{t_0}^{\tau} -f(\theta, x_\theta, \mu_\theta) d\theta - \int_{t_0}^{\tau} c(\theta) \cdot dv_\theta \right\} + \varepsilon E^P(\tau - t_0).$$

By (5.16) and the fact that $P(\tau > t_0) > 0$, we have

$$E^P [W(\tau, x_\tau) - W(t_0, x_0)] > E^P \left\{ \int_{t_0}^{\tau} -f(\theta, x_\theta, \mu_\theta) d\theta - \int_{t_0}^{\tau} c(\theta) \cdot dv_\theta \right\}$$

or

$$W(t_0, x_0) < E^P \left\{ \int_{t_0}^{\tau} f(\theta, x_{\theta}, \mu_{\theta}) d\theta + \int_{t_0}^{\tau} c(\theta) \cdot dv_{\theta} + W(\tau, x_{\tau}) \right\},$$

which contradicts the dynamic programming principle.

The proof of this theorem is therefore complete. $\square$

Let us define the function space

$$\begin{aligned} \mathcal{C}(\Sigma) \equiv \{ & W(\cdot, \cdot) : W \in C(\Sigma; \mathbb{R}) \text{ with } W \text{ bounded and} \\ & |W(t, x) - W(t, y)| \leq C\|x - y\| \text{ for some } C \geq 0\}. \end{aligned}$$

By Theorem 4.4 we know that the value function $W \in \mathcal{C}(\Sigma)$. The proof of the next theorem is a modification of the methods used in Fleming and Soner [5]. For details see Suo [12].

THEOREM 5.5. *There exists a unique viscosity solution in $\mathcal{C}(\Sigma)$ to the dynamic programming equation (5.9) with the boundary condition $W(T, x) = 0$, $x \in \mathbb{R}^d$, which can be identified as the value function.*

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